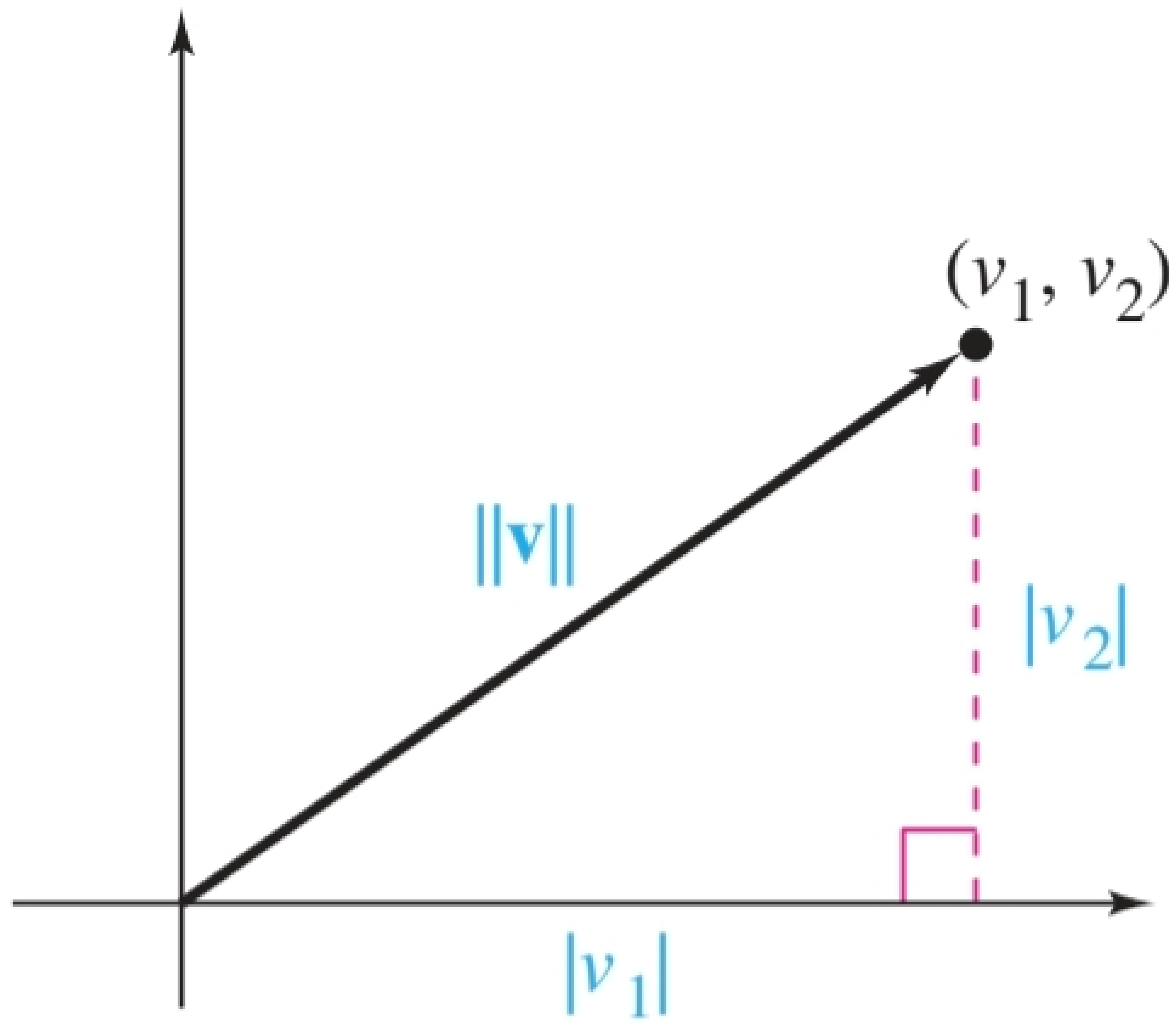
Inner product space

If $v=(v_1,v_2)$ is a vector in $\mathbb{R}^2$, then the length, or norm of v, denoted by $\|v\|$, is the length of the hypotenuse of a right triangle whose legs have lengths of $|v_1|$, and $|v_2|$ as shown below:



$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}$$

Applying pythagorean theorem produces:

$$\|v\|^2 = |v_1|^2 + |v_2|^2 = v_1^2 + v_2^2$$

$$\Longrightarrow \|v\| = \sqrt{v_1^2 + v_2^2}$$

In general:

Definition of the Length of a Vector in R^n

The **length,** or **norm,** of a vector $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ in R^n is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

The length of a vector is also called its **magnitude.** If $\|\mathbf{v}\| = 1$, then the vector $\mathbf{v}$ is a **unit vector.**

This definition shows that the length of a vector cannot be negative. That is, $\|\mathbf{v}\| \geq 0$. Moreover, $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v}$ is the zero vector $\mathbf{0}$.

EXAMPLE 1 The Length of a Vector in R^n

a. In R^5, the length of $\mathbf{v} = (0, -2, 1, 4, -2)$ is

$$\|\mathbf{v}\| = \sqrt{0^2 + (-2)^2 + 1^2 + 4^2 + (-2)^2} = \sqrt{25} = 5.$$

b. In R^3, the length of $\mathbf{v} = \left(2/\sqrt{17}, -2/\sqrt{17}, 3/\sqrt{17}\right)$ is

$$\|\mathbf{v}\| = \sqrt{\left(2/\sqrt{17}\right)^2 + \left(-2/\sqrt{17}\right)^2 + \left(3\sqrt{17}\right)^2} = \sqrt{17/17} = 1.$$

The length of $\mathbf{v}$ is 1, so $\mathbf{v}$ is a unit vector, as shown in Figure 5.2.

Each vector in the standard basis for R^n has length 1 and is a **standard unit vector** in R^n. It is common to denote the standard unit vectors in R^2 and R^3 as

$$\{\mathbf{i}, \mathbf{j}\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad \{\mathbf{i}, \mathbf{j}, \mathbf{k}\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n are **parallel** when one is a scalar multiple of the other—that is, $\mathbf{u} = c\mathbf{v}$. Moreover, if $c > 0$, then $\mathbf{u}$ and $\mathbf{v}$ have the **same direction,** and if $c < 0$, then $\mathbf{u}$ and $\mathbf{v}$ have **opposite directions.** The next theorem gives a formula for finding the length of a scalar multiple of a vector.

THEOREM 5.1 Length of a Scalar Multiple

Let $\mathbf{v}$ be a vector in R^n and let c be a scalar. Then

$$\|c\mathbf{v}\| = |c| \, \|\mathbf{v}\|$$

where $|c|$ is the absolute value of c.

THEOREM 5.2 Unit Vector in the Direction of v

If $\mathbf{v}$ is a nonzero vector in R^n, then the vector

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

has length 1 and has the same direction as $\mathbf{v}$. This vector $\mathbf{u}$ is the **unit vector in the direction of v.**

EXAMPLE 2 Finding a Unit Vector

Find the unit vector in the direction of $\mathbf{v} = (3, -1, 2)$, and verify that this vector has length 1.

SOLUTION

The unit vector in the direction of $\mathbf{v}$ is

$$\frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(3, -1, 2)}{\sqrt{3^2 + (-1)^2 + 2^2}} = \frac{1}{\sqrt{14}}(3, -1, 2) = \left(\frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}\right)$$

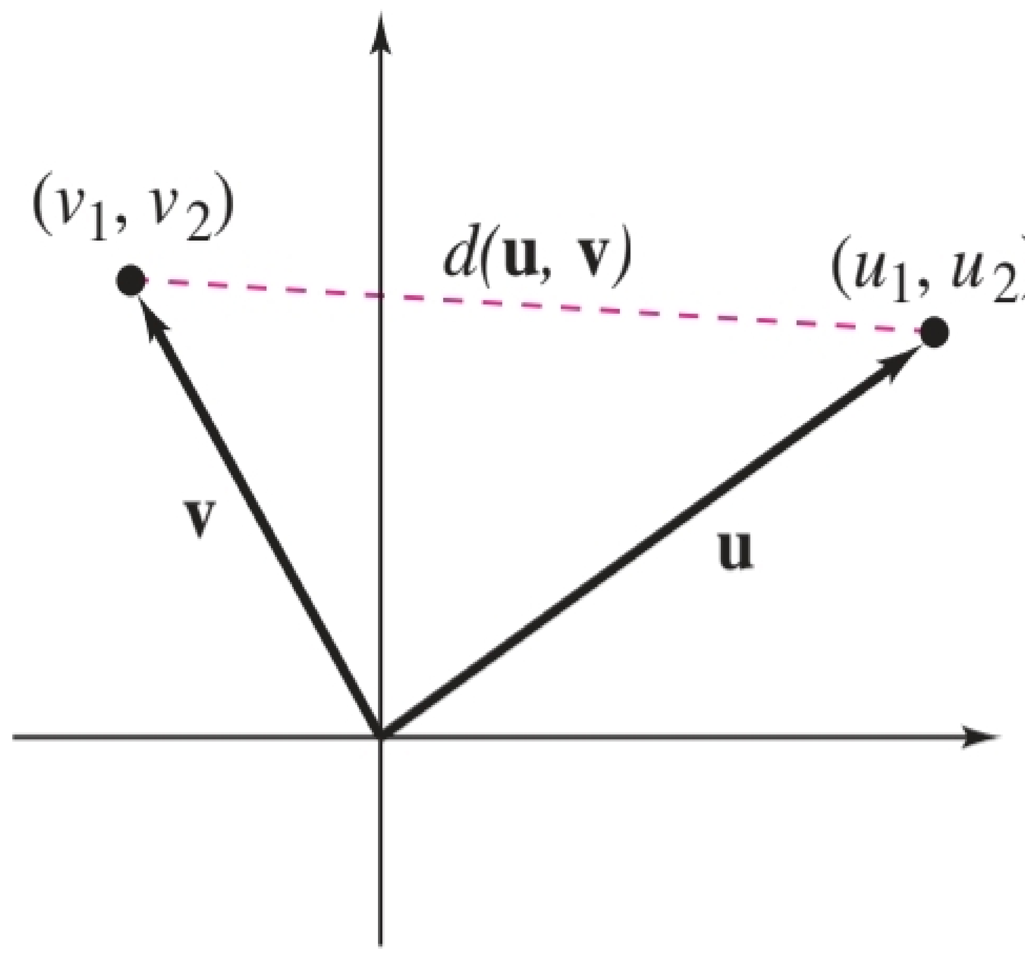
which is a unit vector because

$$\sqrt{\left(\frac{3}{\sqrt{14}}\right)^2 + \left(-\frac{1}{\sqrt{14}}\right)^2 + \left(\frac{2}{\sqrt{14}}\right)^2} = \sqrt{\frac{14}{14}} = 1. \text{ (See Figure 5.3.)}$$

Definition of Distance Between Two Vectors

The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$



$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$

Verify the three properties of distance listed below.

1. $d(\mathbf{u}, \mathbf{v}) \geq 0$
2. $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$.
3. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$

EXAMPLE 3 Finding the Distance Between Two Vectors

a. The distance between $\mathbf{u} = (-1, -4)$ and $\mathbf{v} = (2, 3)$ is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(-1 - 2, -4 - 3)\| = \sqrt{(-3)^2 + (-7)^2} = \sqrt{58}.$$

b. The distance between $\mathbf{u} = (0, 2, 2)$ and $\mathbf{v} = (2, 0, 1)$ is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(0 - 2, 2 - 0, 2 - 1)\| = \sqrt{(-2)^2 + 2^2 + 1^2} = 3.$$

c. The distance between $\mathbf{u} = (3, -1, 0, -3)$ and $\mathbf{v} = (4, 0, 1, 2)$ is

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| \\ &= \|(3 - 4, -1 - 0, 0 - 1, -3 - 2)\| \\ &= \sqrt{(-1)^2 + (-1)^2 + (-1)^2 + (-5)^2} \\ &= \sqrt{28} \\ &= 2\sqrt{7}. \end{aligned}$$

Definition of Dot Product in R^n

The **dot product** of $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ is the *scalar* quantity

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

EXAMPLE 4 Finding the Dot Product of Two Vectors

The dot product of $\mathbf{u} = (1, 2, 0, -3)$ and $\mathbf{v} = (3, -2, 4, 2)$ is

$$\mathbf{u} \cdot \mathbf{v} = (1)(3) + (2)(-2) + (0)(4) + (-3)(2) = -7.$$

THEOREM 5.3 Properties of the Dot Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in R^n and c is a scalar, then the properties listed below are true.

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
3. $c(\mathbf{u} \cdot \mathbf{v}) = (c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v})$
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$
5. $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

EXAMPLE 5 Finding Dot Products

Let $\mathbf{u} = (2, -2)$, $\mathbf{v} = (5, 8)$, and $\mathbf{w} = (-4, 3)$. Find each quantity.

a. $\mathbf{u} \cdot \mathbf{v}$

b. $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$

c. $\mathbf{u} \cdot (2\mathbf{v})$

d. $\|\mathbf{w}\|^2$

e. $\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w})$

SOLUTION

a. By definition, you have

$$\mathbf{u} \cdot \mathbf{v} = 2(5) + (-2)(8) = -6.$$

b. Using the result in part (a), you have

$$(\mathbf{u} \cdot \mathbf{v})\mathbf{w} = -6\mathbf{w} = -6(-4, 3) = (24, -18).$$

c. By Property 3 of Theorem 5.3, you have

$$\mathbf{u} \cdot (2\mathbf{v}) = 2(\mathbf{u} \cdot \mathbf{v}) = 2(-6) = -12.$$

d. By Property 4 of Theorem 5.3, you have

$$\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w} = (-4)(-4) + (3)(3) = 25.$$

e. $2\mathbf{w} = (-8, 6)$, so you have

$$\mathbf{v} - 2\mathbf{w} = (5 - (-8), 8 - 6) = (13, 2).$$

Consequently,

$$\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w}) = 2(13) + (-2)(2) = 26 - 4 = 22.$$

EXAMPLE 6 Using Properties of the Dot Product

Consider two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n such that $\mathbf{u} \cdot \mathbf{u} = 39$, $\mathbf{u} \cdot \mathbf{v} = -3$, and $\mathbf{v} \cdot \mathbf{v} = 79$. Evaluate $(\mathbf{u} + 2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v})$.

SOLUTION

Using Theorem 5.3, rewrite the dot product as

$$\begin{aligned}(\mathbf{u} + 2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v}) &= \mathbf{u} \cdot (3\mathbf{u} + \mathbf{v}) + (2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v}) \\ &= \mathbf{u} \cdot (3\mathbf{u}) + \mathbf{u} \cdot \mathbf{v} + (2\mathbf{v}) \cdot (3\mathbf{u}) + (2\mathbf{v}) \cdot \mathbf{v} \\ &= 3(\mathbf{u} \cdot \mathbf{u}) + \mathbf{u} \cdot \mathbf{v} + 6(\mathbf{v} \cdot \mathbf{u}) + 2(\mathbf{v} \cdot \mathbf{v}) \\ &= 3(\mathbf{u} \cdot \mathbf{u}) + 7(\mathbf{u} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{v}) \\ &= 3(39) + 7(-3) + 2(79) \\ &= 254.\end{aligned}$$

THEOREM 5.4 The Cauchy-Schwarz Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\,\|\mathbf{v}\|$$

where $|\mathbf{u} \cdot \mathbf{v}|$ denotes the *absolute value* of $\mathbf{u} \cdot \mathbf{v}$.

EXAMPLE 7 Verifying the Cauchy-Schwarz Inequality

Verify the Cauchy-Schwarz Inequality for $\mathbf{u} = (1, -1, 3)$ and $\mathbf{v} = (2, 0, -1)$.

SOLUTION

$\mathbf{u} \cdot \mathbf{v} = -1$, $\mathbf{u} \cdot \mathbf{u} = 11$, and $\mathbf{v} \cdot \mathbf{v} = 5$, so you have

$$|\mathbf{u} \cdot \mathbf{v}| = |-1| = 1$$

and

$$\begin{aligned}\|\mathbf{u}\|\,\|\mathbf{v}\| &= \sqrt{\mathbf{u} \cdot \mathbf{u}}\sqrt{\mathbf{v} \cdot \mathbf{v}} \\ &= \sqrt{11}\sqrt{5} \\ &= \sqrt{55}.\end{aligned}$$

The inequality $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\,\|\mathbf{v}\|$ holds, because $1 \leq \sqrt{55}$.

Definition of Orthogonal Vectors

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n are **orthogonal** when

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

EXAMPLE 9 **Orthogonal Vectors in R^n**

a. The vectors $\mathbf{u} = (1, 0, 0)$ and $\mathbf{v} = (0, 1, 0)$ are orthogonal because

$$\mathbf{u} \cdot \mathbf{v} = (1)(0) + (0)(1) + (0)(0) = 0.$$

b. The vectors $\mathbf{u} = (3, 2, -1, 4)$ and $\mathbf{v} = (1, -1, 1, 0)$ are orthogonal because

$$\mathbf{u} \cdot \mathbf{v} = (3)(1) + (2)(-1) + (-1)(1) + (4)(0) = 0.$$

EXAMPLE 10 **Finding Orthogonal Vectors**

Determine all vectors in R^2 that are orthogonal to $\mathbf{u} = (4, 2)$.

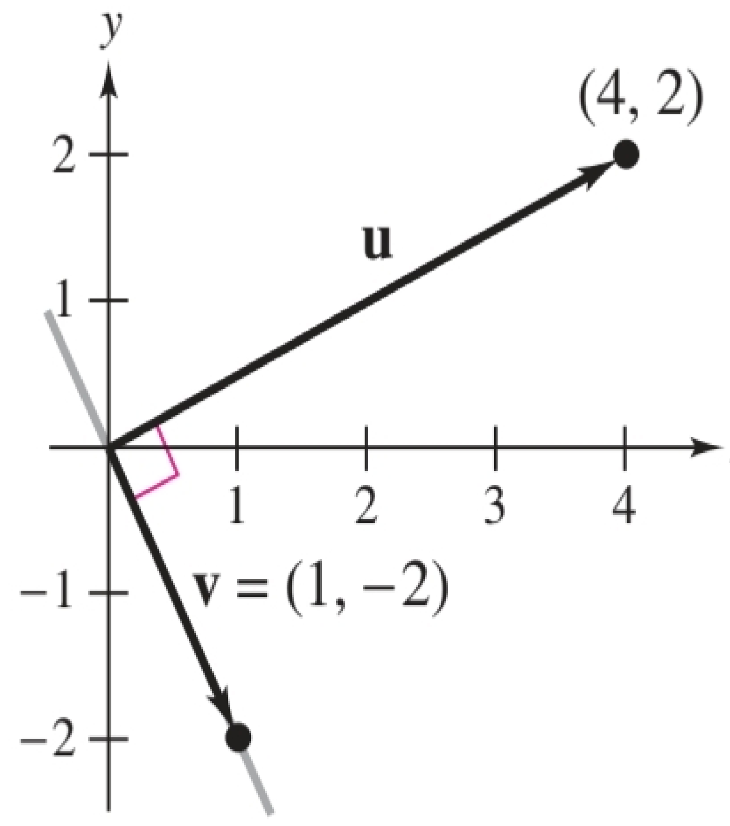
SOLUTION

Let $\mathbf{v} = (v_1, v_2)$ be orthogonal to $\mathbf{u}$. Then

$$\mathbf{u} \cdot \mathbf{v} = (4, 2) \cdot (v_1, v_2) = 4v_1 + 2v_2 = 0$$

which implies that $2v_2 = -4v_1$ and $v_2 = -2v_1$. So, every vector that is orthogonal to $(4, 2)$ is of the form

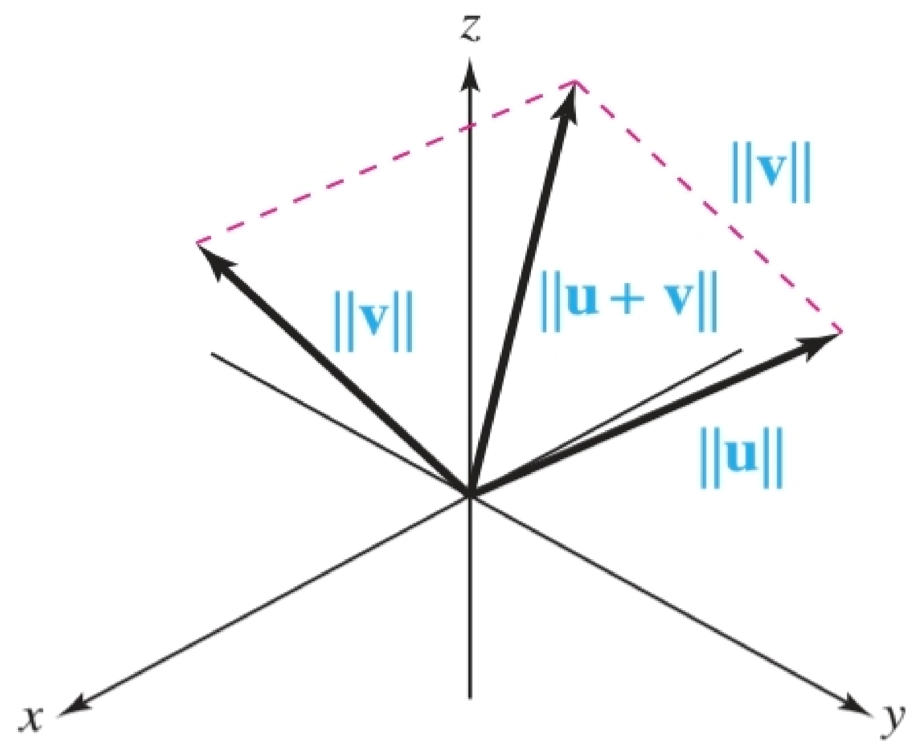
$$\mathbf{v} = (t, -2t) = t(1, -2)$$



THEOREM 5.5 The Triangle Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$



$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

THEOREM 5.6 The Pythagorean Theorem

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

THE DOT PRODUCT AND MATRIX MULTIPLICATION

It is often useful to represent a vector in R^n as an $n \times 1$ column matrix. In this notation, the dot product of two vectors

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

can be represented as the matrix product of the transpose of $\mathbf{u}$ multiplied by $\mathbf{v}$.

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = [u_1 \ \ u_2 \ \ . \ . \ . \ \ u_n]\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = [u_1v_1 + u_2v_2 + \cdot \cdot \cdot + u_nv_n]$$

EXAMPLE 11 Using Matrix Multiplication to Find Dot Products

a. The dot product of the vectors

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

is $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = [2 \;\; 0]\begin{bmatrix} 3 \\ 1 \end{bmatrix} = [(2)(3) + (0)(1)] = 6.$

b. The dot product of the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$$

is $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = [1 \;\; 2 \;\; -1]\begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} = [(1)(3) + (2)(-2) + (-1)(4)] = -5.$

DEFINITION

An **inner product** on a vector space V is a function that, to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V, associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ and satisfies the following axioms, for all $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and all scalars c:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$

2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$

3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$

4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$

A vector space with an inner product is called an **inner product space**.

Example: Fix any two positive numbers say 4 and 5, and for vectors

$u = (u_1, u_2)$ and $v = (v_1, v_2)$ in $\mathbb{R}^2$, set

$\langle u, v \rangle = 4u_1v_1 + 5u_2v_2 \longrightarrow$ (*)

show that equation (*) defines an inner product

sol

Axiom (1) :

$\langle u, v \rangle = 4u_1v_1 + 5u_2v_2 = 4v_1u_1 + 5v_2u_2 = \langle v, u \rangle$

so axiom 1 is satisfied.

Axiom 2: let $w = \langle w_1, w_2 \rangle$

$$\langle u+v, w\rangle = 4(u_1+v_1)w_1 + 5(u_2+v_2)w_2$$
$$= 4u_1w_1 + 4v_1w_1 + 5u_2w_2 + 5v_2w_2$$
$$= (4u_1w_1 + 5u_2w_2) + (4v_1w_1 + 5v_2w_2)$$
$$= \langle u, w\rangle + \langle v, w\rangle.$$

Axiom 3:

$$\langle cu, v\rangle = 4(cu_1)v_1 + 5(cu_2)v_2$$
$$= c(4u_1v_1 + 5u_2v_2) = c\langle u, v\rangle \#$$

Axiom 4:

Note that $\langle u, u\rangle = 4u_1^2 + 5u_2^2 \geq 0$

and $4u_1^2 + 5u_2^2 = 0$ only if $u_1 = u_2 = 0$

and if $u = 0 \Rightarrow \langle u, u\rangle = 0$

So (*) defines an inner product #.

Exercise show: $\langle u, v+w\rangle = \langle u, v\rangle + \langle u, w\rangle$

Example: Let $t_0, t_1, \dots, t_n$ be distinct real numbers

For p and q in $\mathbb{P}_n$, define:

$$\langle p, q\rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \dots + p(t_n)q(t_n) \longrightarrow (*)$$

show that (*) defines inner product #

EXAMPLE 3 Let V be $\mathbb{P}_2$, with the inner product from Example 2, where $t_0 = 0$, $t_1 = \frac{1}{2}$, and $t_2 = 1$. Let $p(t) = 12t^2$ and $q(t) = 2t - 1$. Compute $\langle p, q \rangle$ and $\langle q, q \rangle$.

SOLUTION

$$\begin{aligned}
\langle p, q \rangle &= p(0)q(0) + p\left(\tfrac{1}{2}\right) q\left(\tfrac{1}{2}\right) + p(1)q(1) \\
&= (0)(-1) + (3)(0) + (12)(1) = 12 \\
\langle q, q \rangle &= [q(0)]^2 + [q\left(\tfrac{1}{2}\right)]^2 + [q(1)]^2 \\
&= (-1)^2 + (0)^2 + (1)^2 = 2
\end{aligned}$$

EXAMPLE 4 Let $\mathbb{P}_2$ have the inner product (2) of Example 3. Compute the lengths of the vectors $p(t) = 12t^2$ and $q(t) = 2t - 1$.

SOLUTION

$$\begin{aligned}
\|p\|^2 = \langle p, p \rangle &= [p(0)]^2 + \left[p\left(\tfrac{1}{2}\right)\right]^2 + [p(1)]^2 \\
&= 0 + [3]^2 + [12]^2 = 153 \\
\|p\| &= \sqrt{153}
\end{aligned}$$

From Example 3, $\langle q, q \rangle = 2$. Hence $\|q\| = \sqrt{2}$.

Example: For f, g in $C[a,b]$, set

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt \longrightarrow (*)$$

show that $(*)$ defines an inner product.

sol

Axiom (1)

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt = \int_a^b g(t)\, f(t)\, dt = \langle g, f \rangle$$

Axiom (2) let $h \in C[a,b]$

$$\langle f+g, h \rangle = \int_a^b \big(f(t) + g(t)\big) h(t)\, dt = \int_a^b \big(f(t)h(t) + g(t)h(t)\big) dt$$

$$= \int_a^b f(t)\,h(t)\,dt + \int_a^b g(t)\,h(t)\,dt$$

$$\langle f, h\rangle + \langle g, h\rangle \quad \#$$

Axiom 3 $\langle cf, g\rangle = \int_a^b cf(t)\,g(t)\,dt$

$$= c\int_a^b f(t)\,g(t)\,dt = c\,\langle f, g\rangle \quad \#$$

For Axiom 4, observe that

$$\langle f, f\rangle = \int_a^b [f(t)]^2\,dt \geq 0$$

The function $[f(t)]^2$ is continuous and nonnegative on $[a, b]$. If the definite integral of $[f(t)]^2$ is zero, then $[f(t)]^2$ must be identically zero on $[a, b]$, by a theorem in advanced calculus, in which case f is the zero function. Thus $\langle f, f\rangle = 0$ implies that f is the zero function on $[a, b]$. So (5) defines an inner product on $C[a, b]$. ■